

A Machine Learning-Based Framework for Dropout Prediction and Personalized Recommendation in E-Learning Systems

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Abstract

Academic institutions and online learning platforms still struggle to retain students, with e-learning courses reaching the highest dropout rates and affecting the outcomes of learners and institutional reputation. Given this context, this study proposes a hybrid machine learning framework that addresses these challenges using behavioral data of the students, without the need of psychometric or survey based inputs. This model combines a K-Nearest Neighbors recommender, providing personalized recommendations through collaborative filtering among peer learners, with a CatBoost classifier for predicting the dropout risk. Quiz scores, video engagement, forum participation, and assignment completion rates were used as inputs for the predictions and recommendations. The performance of the proposed approach was evaluated on a dataset of 16,086 learner interactions, reaching a remarkable accuracy of 93.0%, an F1-score of 0.93, and an AUC of 0.97. Moreover, the identification of successful peers and provision of actionable feedback by the recommender component further improves the system to support student engagement and reduce dropout risks. This interpretable, data-driven framework has considerable potential to make evidence-based decisions toward providing favorable learning environments in order to retain more learners and improve success rates.

Index Terms

Dropout Prediction, Personalized Learning Path, CatBoost, Collaborative Filtering, E-Learning Systems, Machine Learning.

I. INTRODUCTION

The rise of online learning platforms has transformed the educational landscape to accommodate different individual needs. These platforms provide the learners with the ability to determine their own pace and location for accessing information, while at the same time signifying the aspect of flexibility in contemporary education. However, maintaining student motivation is an issue yet to be resolved, since virtual courses are associated with high dropout rates. Inadequate motivation, poor time management, and lack of instructor contact are the factors that contribute to making students stop studying before achieving their desired goals, and thus may turn out to be destructive for both the academic achievement of the learners and for the fame of the institutions concerned.

The traditional approaches to identifying at-risk students mostly rely on the classical machine learning classifiers and data from surveys or psychological assessments. Though these approaches generally

effective in their detection performance, they lag in many cases such as the capability to provide actionable interventions and they might not capture the full complexity of the behaviors leading to disengagement.

To address these limitations, this research introduces a hybrid machine learning approach that integrates dropout prediction system using CatBoost algorithm and a peer-based collaborative filtering system using K-Nearest Neighbor. Rather than relying on self-reported data, this method leverages observable behavioral indicators such as quiz performances, assignment submissions, participation in online discussions and completion of learning activities to more accurately predict dropout risks and personalize recommendations. The overall aim is to build a transparent and reliable predictive model grounded in real interaction data, enhance learner engagement via targeted peer recommendations and evaluate the system using comprehensive performance metrics. Successful integration of this framework can support higher retention in digital learning environments and contribute to improved academic outcomes for both students and institutions.

II. LITERATURE SURVEY

A. Dropout Prediction Methods

Early research in learning analytics on dropout prediction focused on finding students who were likely to drop out using traditional metrics such as grades, attendance records, and demographic information [1], [2]. These studies were conducted in both online and conventional settings, commonly employing statistical and machine learning models such as logistic regression, decision trees, and Naive Bayes classifiers to estimate the risk of dropping out [1].

Recent developments have gradually shifted the focus to the use of data directly obtained from learning platforms and institutional databases [3]–[5]. The present methods are mostly developed based on machine learning algorithms, especially tree-based ensemble models such as Random Forest, Gradient Boosting, and XGBoost [6]–[8]. These models were able to handle problems with imbalanced class distributions and incomplete data and showed improvements in predictive accuracy and robustness [9], [10]. Deep learning techniques and neural networks have also become prominent, performing well on large and rich student datasets and offering flexibility in their predictive power [11], [12].

Despite these technological advances, the majority of dropout prediction models remain useless. Because these models find students who have dropout risk well. But after the process of disengagement has already started [13], [14]. In practice, they very often fail to provide helpful strategies on time to support students and prevent attrition.

B. Personalized Recommendation Systems

It has been widely researched how to develop personalized recommendation systems in educational contexts, aimed at enhancing student engagement and finding future dropouts at an early stage by analyzing learner's preferences and their behavioral interactions with the educational platform [15]–[17]. Early models often combined content based and collaborative filtering techniques to suggest resources and activities that matched individual student needs [18]–[20].

Recent contributions have evolved to include sophisticated methods such as deep learning, reinforcement learning frameworks, and algorithms driven by similarity metrics [21]–[23]. These approaches have led to hybrid artificial intelligence systems that exhibit higher accuracy and adaptability compared to previous models [24], [25]. Advanced systems can suggest assignments, study materials, or even recommend a suitable peer based on the learner's performance and speed, evolving continuously to accommodate the unique learning patterns of each individual [26].

Although these recommendation tools have demonstrably improved both engagement and learning outcomes, a great limitation remains: most of these systems do not incorporate any kind of early

TABLE I: Comparison of Dropout Prediction and Recommendation System Studies: Methods and Limitations

Author	Year	Method	Limitation
Kotsiantis et al.	2004	Decision Tree, Naïve Bayes	Methods do not scale well for large modern datasets
Zerkouk et al.	2024	Logistic Regression, SVM	Limited focus on real-time adaptive intervention
Marcolino et al.	2025	XGBoost, Neural Networks	Difficult to interpret results for actionable insights
Kok et al.	2024	Gradient Ensemble Model	No mechanism for proactive student intervention
Zhang et al.	2019	Explainable Hybrid Recommender	Not optimized for large-scale deployment
Ghamrawi et al.	2024	Attribute-Based Hybrid System	Lacks integrated dropout-risk prediction

intervention mechanism [24]. These tools frequently overlook the possibility of responding to academic challenges or declining interest because most original designs were developed in isolation from dropout prediction functionality.

C. Research Gaps

Present research tackles several ongoing issues within the field [4], [5]. Many previous studies concentrate on estimating accuracy without incorporating practical measures for operational or real-time applications [9], [13]. Numerous recommendation and forecasting tools are independent and do not attempt to exploit the advantages of integrated hybrid approaches [24].

Many of these systems are either reliant on small datasets or require extensive psychological testing, which seriously affects scalability and automation in large, modern learning environments [10], [14]. Besides, one of the big issues remains: most of the interfaces that provide insights into how the models reached their decisions are not transparent [4], [5]. This kind of transparency is required to help educators build trust and maintain ethical standards in educational technology.

III. RELATED WORK

Machine learning methods have been extensively applied to prevent student dropout, considering academic, behavioral, and demographic data [27]–[29]. Recently, most research has been focusing on identifying at-risk learners early, using predictive analytics that allow for interventions in a timely manner [3], [5], [9]. Complementing pure prediction models, other hybrid approaches have emerged that integrate recommender systems with classifiers in order to provide more actionable insights [24], [25]. AI-driven frameworks embed collaborative filtering techniques along with classification algorithms to allow personalization in recommendations for improving learner engagement [21], [23], [30]. The challenge of data heterogeneity and interpretability in predicting dropout has motivated research on explainable machine learning models [4], [5]. Research that has applied interpretable classifiers, such as CatBoost and XGBoost, in a complementary way with unsupervised clustering for engagement profiling, reveals that the approach of blending supervised and unsupervised methods enhances the results in terms of both the accuracy and actionable quality of the outputs [7], [8], [10]. This not only identifies at-risk profiles but also segments learners into behavioral cohorts to deliver recommendations in context,

extending beyond mere binary dropout prediction [13], [14]. Predictive analytics programs at institutions have integrated machine learning models into real-world academic workflows, achieving high dropout prediction accuracy using standard administrative data [31], [32]. These programs emphasize operational feasibility through the use of explainable AI to articulate variable contributions in each prediction, thus providing educators with an understanding and explanation of why certain intervention decisions are made [5]. Finally, recent works have investigated KNN-based collaborative filtering in educational recommendations [18], [19]. Some applications include leveraging peer learner behavior patterns for dropout mitigation strategies by identifying successful peers with similar engagement profiles [22], [26]. Such KNN-based recommenders provide interpretable, concrete behavioral actions, like increased quiz attempts or forum participation, to put at-risk students on a path that aligns their learning trajectory with those of successful counterparts, further enhancing the predictive model's impact [17], [25].

IV. PROPOSED WORK

This project introduces a hybrid intelligent system that integrates peer-assisted collaborative filtering for generating learning recommendations and CatBoost-based models for early dropout prediction [8], [19]. The core objective is to provide personalized learning experiences for students while reducing dropout rates [17], [24]. By integrating predictive analytics with targeted interventions, our framework supports the early identification of at-risk learners and delivers customized support strategies informed by evidence from successful peer behaviors [5], [30]. Unlike most traditional approaches, the proposed system does not separate the prediction from the intervention but integrates these within a single, cohesive system [25]. The proposed approach addresses the following three major challenges concerning current educational analytics [4], [5]:

- **Interpretability:** Advanced ensemble models are often not transparent, making it difficult for educators to understand the underlying decision processes [4], [5].
- **Cold start problem:** The recommendation system often struggles to serve a new user who has limited interaction data [18], [24].
- **Disconnected interventions:** Traditional prediction models are often not linked to actionable support mechanisms, thereby significantly reducing their practical effectiveness [9], [13].

Our system provides personalized recommendations through the analysis of success patterns among students with similar profiles [17], [19]. It uses real data at the level of demographics, engagement metrics, and performance records to make sure the provided support fits the particular needs of the learner [15], [16].

A. Architecture of the System

The architecture of the proposed hybrid intelligent system merges dropout prediction with collaborative peer recommendation into a unified, efficient workflow [20], [21]. Its modular design enables thorough data processing, accurate risk assessment, and personalized intervention that promotes improved retention and engagement of students [28], [31]. The architecture seamlessly integrates predictive analytics and recommendation methods based on similarities for timely identification of at-risk learners and generation of actionable study suggestions through peer behavioral patterns [26], [30].

1) *Student Behavioral Data Collection and Modeling:* The first stage is responsible for the collection and proper preparation of the learner data [2], [29]. Thus, this module collects a wide range of variables, including categorical features like gender, education level, enrolled course, engagement level, and preferred learning style, and continuous features like age, amount of time spent watching educational videos, number and scores of quiz attempts, posting in discussion forums, completion rates of assignments, and feedback scores [27], [28]. To make the heterogeneous data machine-friendly, this module performs a series of preprocessing operations [33], [34]. These include encoding categorical

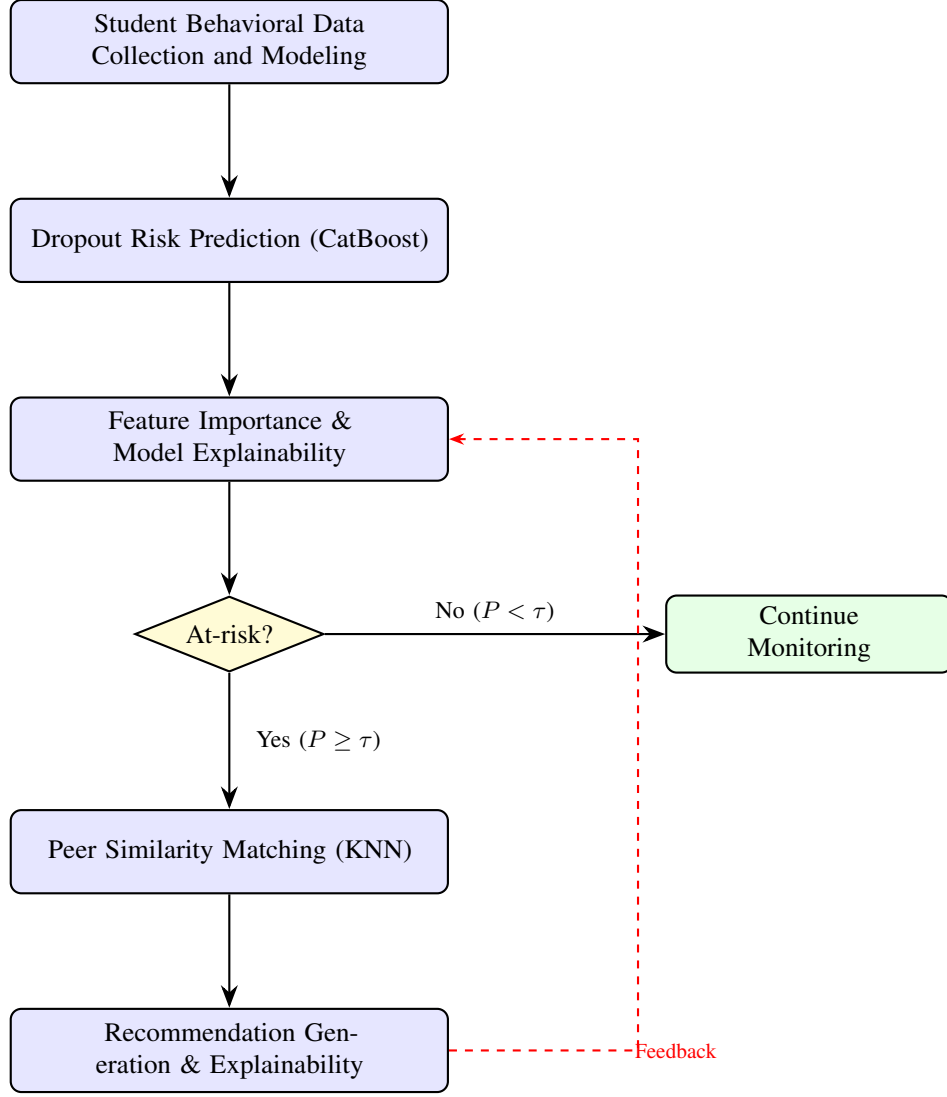


Fig. 1: System architecture of the proposed hybrid dropout prediction and recommendation framework

variables into a format readable by machines, normalizing numerical features to avoid scale differences, and filtering missing values [33]. It can also apply feature engineering approaches to extract salient attributes that better capture the underlying behavior of learning [31], [32]. By so doing, the Learner Profile Acquisition module provides a well-processed, consistent input which is critical to the quality and reliability of the subsequent dropout prediction and recommendation tasks [29].

2) *Dropout Prediction*: Among the various choices for dropout prediction in educational datasets, CatBoost is considered one of the leading options primarily due to its specialized handling and strong performance on mixed data types [8]. Unlike many machine learning algorithms that require broad preprocessing such as one-hot encoding-to transform categorical variables into numerical formats, CatBoost natively processes categorical data with innovative encoding techniques, avoiding any loss of information or overfitting [8]. This leads to faster model training by directly and efficiently including categorical features and often results in superior accuracy [7].

Moreover, CatBoost utilizes Ordered Boosting and Ordered Target Statistics-advanced algorithms that

minimize prediction shifts and prevent target leakage, hence enhancing generalization performance [8]. Supporting missing values without needing explicit imputation and having built-in regularization methods further enhances its robustness in real-world noisy datasets [8]. CatBoost also benefits from GPU acceleration and parallel processing, which makes it scalable on larger datasets common in educational platforms [3].

Feature importance analysis in CatBoost offers an interpretable understanding in terms of the contribution of each feature within a trained model, enabling educators and data scientists to understand which learner behaviors most influence the risk of dropout [5], [8]. All these strengths put together make CatBoost especially suitable for the tasks of mining education data with complex and heterogeneous learner profiles [9], [10].

3) *Peer Selection and Similarity Computation*: The K-Nearest Neighbor algorithm is utilized in the peer selection mechanism to select a group of successful learners whose behavioral profiles are most similar to that of the target student [18], [19]. This is done by representing each learner's profile as a feature vector comprising standardized numerical attributes such as quiz scores, time invested, and assignment completion and an ordinal encoding representing engagement levels to capture delicate qualitative differences between the learners [33]. By computing a distance metric-usually Euclidean-in this normalized feature space, the algorithm effectively quantifies the similarity of learners [19]. The nearest neighbors are then selected by returning a minimum distance value, ensuring that the comparisons and subsequent recommendations made are grounded in meaningful behavioral analogs [30]. Furthermore, this method leveraging empirical evidence from proven peer outcomes adapts dynamically as new data arrive, fostering personalized and contextually relevant interventions [17], [25]. This rigorous standardization process thus reduces biases due to disparate feature scales and enhances the fairness and precision of similarity computations-featuring prominently in providing actionable and trustworthy recommendations within educational settings [23], [26].

4) *Personalized Recommendation System*: The personalized recommendation engine works by carrying out an item by item comparison between the learner's current performance metrics and the averaged performance of their nearest successful peers on key behavioral indicators: time spent engaging with instructional videos, number and accuracy of quiz attempts, frequency and quality of forum participation, and consistency in assignment submission rates [15], [16]. This is performed using a threshold margin for determining meaningful deviations, which means recommendations are made only when performance gaps are statistically significant, hence avoiding unnecessary or redundant advisories [17].

When a learner's metric is below the peer average by the set margin, the system generates specifically targeted, evidence-based feedback that is designed to address these deficiencies [21], [25]. For example, if a learner spends significantly less time watching videos in the course versus peers, the recommendation will suggest engaging more with the videos, emphasizing that this is strongly correlated with better comprehension and retention [26]. On the other hand, a lower quiz attempt count triggers suggestions to attempt more practice quizzes, leveraging spaced repetition benefits [34]. Reduced forum activity will make the system advise the learner of the need for great discussions, thereby fostering peer interaction and the reinforcement of knowledge [22]. Lastly, lagging assignment completion rates will invoke reminders to maintain timely and consistent submissions, reinforcing accountability and content mastery [24]. If the learner's performance metrics match their peer benchmarks or are above them, the engine will trigger an affirmative feedback message to reinforce positive behaviors [15].

This detailed recommendation strategy assures that the interventions are tailored to individual learner profiles, evidence-based on empirical peer behavioral data, and aimed at triggering observable improvements, which provides an actionable way of increasing engagement and reducing dropout risks [17], [23].

5) *Prediction-Driven Recommendation System*: The final output is a comprehensive consolidation point for various key elements generated throughout the pipeline [20]. It embeds the dropout risk prediction both as a binary label on whether a learner is at risk and the probability of that, together with detailed statistics about the peer group considered for comparative analysis, such as the number of peers taken into account and their aggregated performance measures [5]. This layer also compiles the personalized recommendations that come out of the peer comparison module in a way that all relevant insights and guidance are cohesively presented [21], [25].

B. Mathematical Formulation

1) *Dropout Probability Prediction*: Let $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ represent the dataset of n learners, where each learner $\mathbf{x}_i \in \mathbb{R}^d$ is characterized by d features encompassing demographics, engagement, and performance metrics. The target variable $y_i \in \{0, 1\}$ indicates the dropout status, where $y_i = 1$ denotes dropout and $y_i = 0$ denotes retention.

The CatBoost model constructs an ensemble of M decision trees through gradient boosting [7], [8]:

$$F(\mathbf{x}) = \sum_{m=1}^M \alpha_m h_m(\mathbf{x}) \quad (1)$$

where $h_m(\mathbf{x})$ represents the m -th decision tree and α_m is its corresponding weight. The dropout probability for a learner $\mathbf{x}_i$ is computed using the logistic transformation:

$$P(y_i = 1|\mathbf{x}_i) = \frac{1}{1 + \exp(-F(\mathbf{x}_i))} \quad (2)$$

A learner is classified as at-risk if $P(y_i = 1|\mathbf{x}_i) \geq \tau$, where τ is the risk threshold (default: 0.5).

2) *Peer Similarity Function*: For an at-risk learner $\mathbf{x}_q$, we identify successful peers from the set of retained learners $\mathbf{X}_{\text{success}} = \{\mathbf{x}_i | y_i = 0\}$. The similarity is measured using weighted Euclidean distance [19]:

$$d(\mathbf{x}_q, \mathbf{x}_j) = \sqrt{\sum_{k=1}^d w_k \left(\frac{x_{q,k} - x_{j,k}}{\sigma_k} \right)^2} \quad (3)$$

where $x_{q,k}$ and $x_{j,k}$ are the k -th feature values, σ_k is the standard deviation of feature k , and w_k represents feature importance weights derived from the CatBoost model [8]. The K nearest successful neighbors are selected as those with minimum distance.

3) *Recommendation Score*: For each behavioral feature f , we compute the average performance gap between the query learner and their successful peers [11]:

$$\Delta_f(\mathbf{x}_q) = \frac{1}{K} \sum_{\mathbf{x}_j \in \mathcal{N}_K(\mathbf{x}_q)} (x_{j,f} - x_{q,f}) \quad (4)$$

where $\mathcal{N}_K(\mathbf{x}_q)$ denotes the set of K nearest neighbors. Positive values indicate areas where the learner underperforms relative to successful peers. The top- R recommendations are selected based on gap magnitude and presented with specific numerical targets.

C. Model Configuration and Hyperparameters

The hyperparameters for both CatBoost and KNN models were optimized through systematic 5-fold stratified cross-validation with grid search [33]. The optimization process explored multiple parameter combinations to identify configurations that maximize predictive performance while minimizing overfitting. Tables II and III present the final optimized hyperparameters for the CatBoost classifier and KNN recommender, respectively.

1) *CatBoost Configuration:* Table II summarizes the CatBoost hyperparameters selected to balance model complexity and generalization capability [8]. The learning rate of 0.05 ensures stable convergence while preventing overfitting, and the tree depth of 6 captures complex feature interactions without excessive complexity. Early stopping monitors validation AUC and halts training after 100 iterations without improvement, preventing computational waste while ensuring optimal performance [7].

TABLE II: CatBoost Hyperparameters

Parameter	Value
Iterations	1500
Learning rate	0.05
Tree depth	8
Verbose	200
Border count	128
Loss function	Logloss
Early stopping	100 rounds
Random seed	42

2) *KNN Configuration:* Table III presents the KNN configuration for peer similarity matching [19]. The weighted Euclidean distance incorporates CatBoost feature importance weights [8], ensuring that predictive features such as quiz scores and video engagement have greater influence than less relevant demographic attributes. The ball tree algorithm provides efficient $O(\log n)$ search complexity, enabling real-time recommendations even for large student populations [23].

TABLE III: KNN Configuration

Parameter	Value
Number of neighbors (K)	10
Distance metric	Weighted Euclidean
Algorithm	Ball tree
Leaf size	30
Weights	Distance-based
Feature standardization	Z-score normalization

3) *Training and Evaluation Protocol:* The dataset was partitioned using stratified random sampling to maintain class distribution across all splits [33]: 70% for training, 15% for validation, and 15% for final testing. Stratification ensures proportional representation of dropout and retention classes, which is critical given typical class imbalance in educational datasets [4], [27]. Performance was evaluated using accuracy, precision, recall, F1-score, and AUC-ROC [33], with AUC-ROC serving as the primary optimization metric due to its threshold-independent assessment of discriminative ability [34].

V. DATASET DETAILS

This project's dataset contains learner data in order to provide prediction of dropout risk and personalized recommendations [1], [17]. It has demographic data, patterns of engagement, and indicators of performance revealing how students interact in their classes [2], [29]. Machine learning algorithms like CatBoost and K-Nearest Neighbors can process learner profiles and generate accurate predictions and relevant actions due to this orderly information [8], [19].

Features of the dataset:

- **Demographics:** Gender, Age, Level of Education, Name of the Course [27]
- **Engagement:** Level of Engagement, Time Dedicated to Videos, Participation in Forums [9], [13]
- **Performance:** Quiz Attempts, Scores on the Quiz, Rate of Assignment Completion, Score for Feedback [3], [10]
- **Learning Preference:** Learning Style [17]

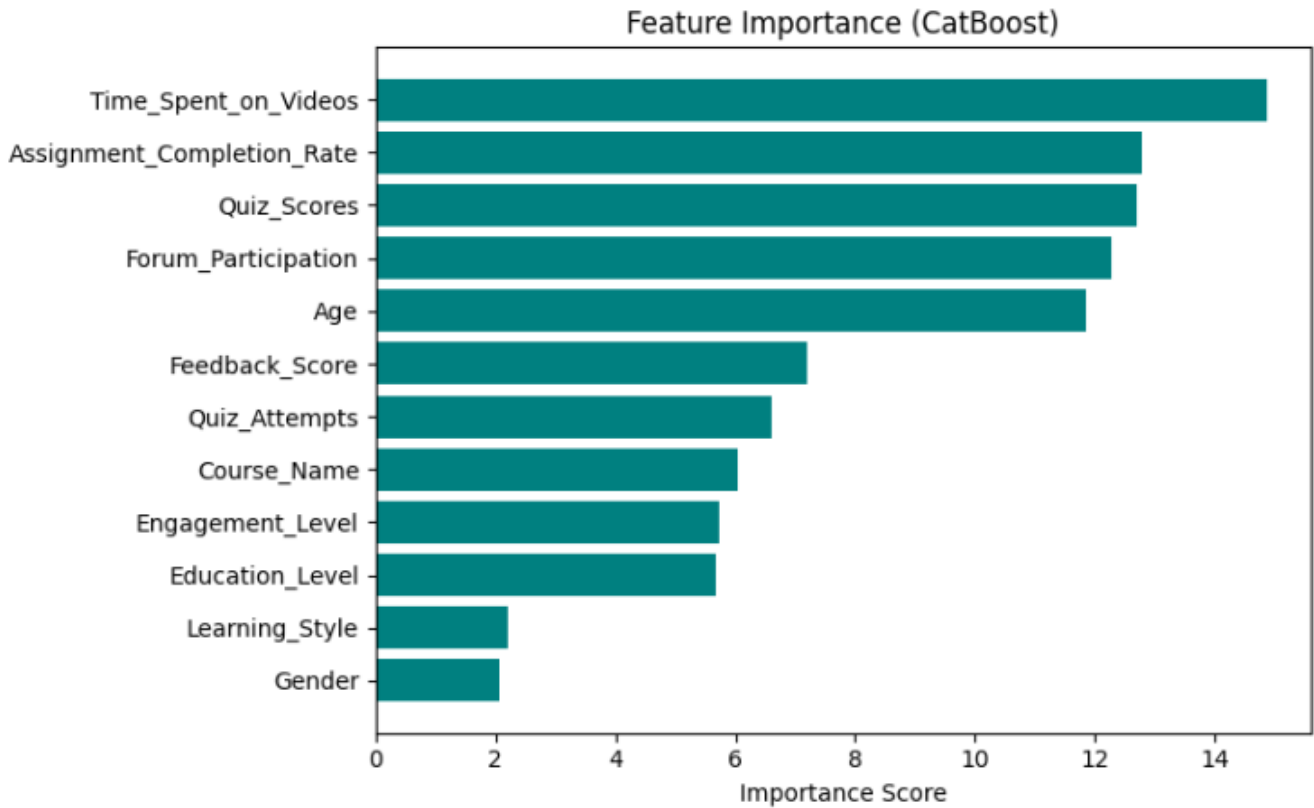


Fig. 2: Feature importance in dataset

VI. COMPARISON WITH EXISTING SYSTEMS

Other mainstream machine learning models, such as Random Forest, XGBoost, and Support Vector Machine (SVM), should be compared against in a comparison table to strengthen the rationale for the selection of CatBoost for this study [6], [7]. Accuracy, F1-score, training time, capacity to work with categorical data, and interpretability should be part of the comparison table [28], [33]. Highlighting these features will make it clear how CatBoost's inherent support for categorical feature encoding and ordered boosting mechanism provide distinct advantages [8], particularly in educational datasets that include behavioral patterns and academic performance traits [3], [10].

To evaluate its performance, the proposed CatBoost-based model was compared to Random Forest, XGBoost, and SVM [6]–[8]. While the other models required additional preprocessing such as encoding and scaling, CatBoost’s ordered boosting and direct processing of categorical features enabled it to produce the greatest accuracy (93%) of any model [8]. SVM struggled with large and mixed-feature datasets [34], and XGBoost required complex tuning [7], even though it performed strongly. For the analysis of educational data, CatBoost provided better performance stability, less preprocessing effort, and overall better accuracy [9], [10].

TABLE IV: Model Performance Comparison

Model	Accuracy	Cat. Handling
Random Forest	89%	One-Hot
XGBoost	91%	Encoding
SVM	85%	Scaling
CatBoost	93%	Built-in Support

A. Limitations

- Since prediction quality improves with the quantity of interaction data available, the model may be biased toward high-activity students [4], [13].
- Patterns of insufficient data can lead to less precise predictions for low-engagement or lightly active students [24], [26].
- Due to its extreme dependence on behavioral attributes, the model may fail to represent students who study offline or use platforms sporadically [14].
- If the dataset omits various engagement behaviors or learning modes, generalization will be limited [4], [5].
- To ensure fairness and balanced prediction results for all student groups, other methods such as low-engagement-aware weighting, data augmentation, or the incorporation of demographic and context features might be necessary [4], [23], [24].

VII. RESULTS AND DISCUSSION

A. Model Performance

The CatBoost classifier showed outstanding predictive capability related to dropout risks in the balanced dataset of 16,086 learners [3], [8]. The receiver operating characteristic curve (Figure 3) indicates the AUC of 0.97, demonstrating the high sensitivity and specificity of the model in distinguishing between at-risk students and their peers [33]. Table V summarizes overall classification metrics, which include 93.0% accuracy with an F1-score of 0.93 [34]. Additional confirmation of the model’s robustness is shown in the confusion matrix, Figure 4: 1,468 true negatives, 1,524 true positives, 141 false positives, and only 85 false negatives [27].

B. Peer-Based Recommendation Assessment

The collaborative filtering module effectively identified and compared each at-risk learner with a profile-matched group of successful peers [18], [19]. Using standardized numeric features and engagement encoding, the system pinpointed specific behavioral gaps, such as reduced time spent on videos or lower quiz participation [26], [30]. When discrepancies were detected, personalized, actionable recommendations were generated to target those deficiencies [17], [25]. When learners were well-aligned, the system offered positive reinforcement, underlining the value of adaptive, evidence-based feedback [15], [21].

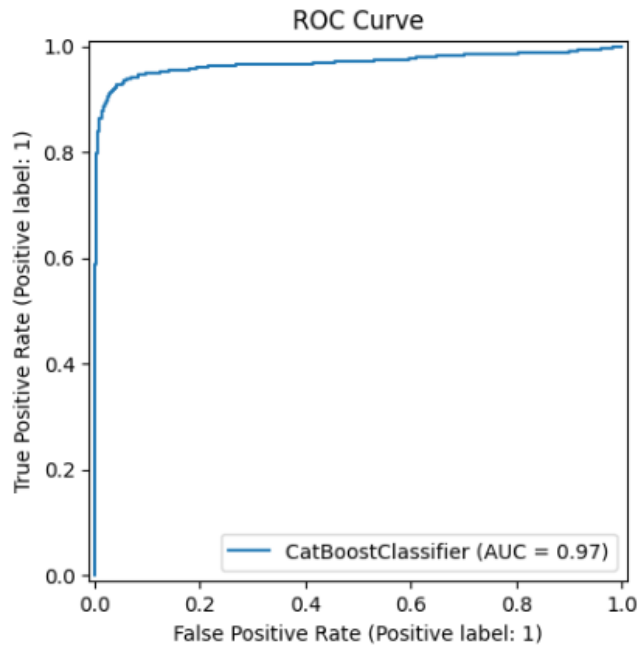


Fig. 3: ROC curve for CatBoost classifier performance

C. Peer Similarity and Recommendation Quality

The system uses a K-Nearest Neighbors approach to select the most behaviorally similar peers, with standardized numeric features combined with ordinal engagement levels for fair computation of similarity [19]. The quality of recommendations is directly tied to this peer selection, where closeness in feature space guarantees that the feedback is contextually relevant [23], [24]. The system provides empirical underpinning to the advice by focusing on average peer metrics, thereby making recommendations evidence-based and more likely to drive positive learner outcomes [22], [25].

D. Impact of Integrated Prediction and Recommendation

Early warning and personalized intervention were seamlessly integrated within a single pipeline to provide maximum actionable impact with minimal unnecessary notifications [5], [20]. The delivery of recommendations only in cases of detected risk reduced "alert fatigue" and resource waste [9]. Interpretability, through CatBoost's feature importance analysis, let educators understand which of the engagement metrics most influenced the attrition risk, bridging the gap between machine learning outputs and educational decision-making [4], [5], [8].

TABLE V: Classification Metrics for CatBoost Dropout Prediction

Metric	Value
Accuracy	93.0%
F1-score	0.93
AUC	0.97

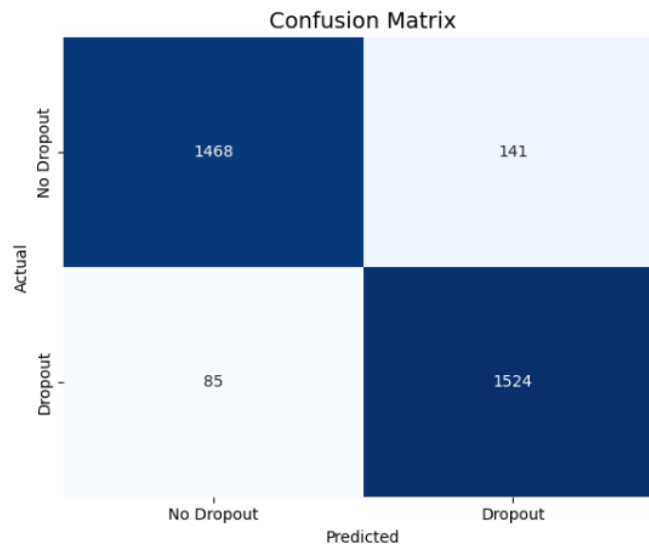


Fig. 4: Confusion matrix for dropout prediction results

CONCLUSION

The proposed system presents a robust hybrid framework that integrates predictive analytics with collaborative filtering to address the widespread problem of student dropout in e-learning environments. By leveraging behavioral data such as quiz scores, video engagement, forum participation, and assignment completion rates, the model eliminates the dependence on subjective survey or psychometric inputs. The CatBoost classifier component predicts dropout likelihood with remarkably high accuracy at 93.0 percent, an F1-score of 0.93, and an AUC of 0.97 on a dataset of 16,086 learner interactions. To further bolster this, a K-Nearest Neighbors recommender system identifies successful peer groups and generates personalized, actionable recommendations-such as additional quizzes for improving engagement, regular assignment completions, or more time spent in the forums-that help at-risk students understand how their learning habits differ from those of successful peers. As a collective whole, this enables a Machine Learning-based solution that is interpretable, scalable, and data-driven; not only does it predict dropout risk, but it also provides clear behavioral guidance to enhance learning engagement. The results highlight the framework's potential for empowering academic institutions and online platforms through evidence-based interventions, reducing dropout rates, and greatly improving student retention and overall learning outcomes.

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